

# TP1 : TypeScript

## Problème : Gestion d'une plateforme de cours en ligne (E-learning)

### 1°) Définition des types de base

a.

```
export type Role = "student" | "teacher" | "admin";
```

b.

```
export interface User {  
    id: number;  
    name: string;  
    role: Role;  
    email?: string;  
}
```

### 2°) POO

a.

```
export abstract class Course {  
    title : string ;  
  
    abstract getDescription(): string;  
}
```

b.

```
export class VideoCourse extends Course {  
    duration : number ;  
    constructor(title: string,duration: number) {  
        super(title);  
        this.title=title ;  
    }  
}
```

```
export class LiveCourse extends Course {  
    date : Date ;  
    constructor(title: string,date: Date) {  
        super(title);  
        this.date=date  
    }  
}
```

### 3°) Interfaces et Génériques

a.

```
export interface Repository<T> {  
  add(item: T): void;  
  remove(id: number): void;  
  getAll(): T[];  
}
```

b.

```
export default class UserRepository implements Repository<User> {  
  private items: User[] = [];  
  
  add(item: User): void {  
    if (this.items.some(u => u.id === item.id)) {  
      throw new Error(`User with id ${item.id} already exists`);  
    }  
    this.items.push(item);  
  }  
  
  remove(id: number): void {  
    this.items = this.items.filter(i => i.id !== id);  
  }  
  
  getAll(): User[] {  
    return [...this.items];  
  }  
  
  findById(id: number): User | undefined {  
    return this.items.find(u => u.id === id);  
  }  
}
```

### 4°) Types avancés